CATEGORY THEORY ABSTRACTIONS FOR tnreason

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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ABSTRACT

We investigate the abstractions of tensor networks by string diagrams of categories for usage in the tnreason formalism.

1 Semiring contractions as categories

Each semiring $\mathbf{S} = (S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ defines a category:

- Objects are the elements of the cyclic group
- Morphisms are maps from the objects to $\mathbf{S}$

Open question: Are monads the property needed for message passing?

2 Normalization

Directed tensors are normalizations in the Mat category.

When mapping the Hilb elements to Quant, quantum states are the normalized pure density matrices and unitaries channels.

3 Encoding schemes as functors

- Basis encodings, selection encodings as functors from the FinSet category to the Mat category
- Computation circuits and activation circuits as functors from the FinSet category to the Hilb category

References